QUANTUM CIRCUIT SIMULATION

VQETape

Exact VQE program co-design with an audited same-node TensorCircuit-NG baseline

8.2%	28.3%	1.27x
faster amortized objective	less host peak RSS	slower best VQ warm
VQ spatial vs TC-NG	473.9 vs 661.3 MiB	3.379 vs 2.658 ms

HONEST VERDICT: PARTIALLY MET

VQETape spatial transfer is 8.2% faster on the declared compile + first + 100-warm objective and uses 28.3% less host peak RSS than TensorCircuit-NG on the same RTX 3090. TensorCircuit-NG still owns the fastest warm kernel, sampled GPU memory is tied, and the formal N=32,L=16 Fig. 2 run is absent.

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Challenge #33 - PR #263
Evidence snapshot: 748c5e974573
30 July 2026

1. Challenge and evaluation contract

The requested system should iterate over exact VQE implementations, auto-evaluate them, and beat the TensorCircuit-NG official baseline in space and time - including compilation, first execution, and subsequent warm execution.

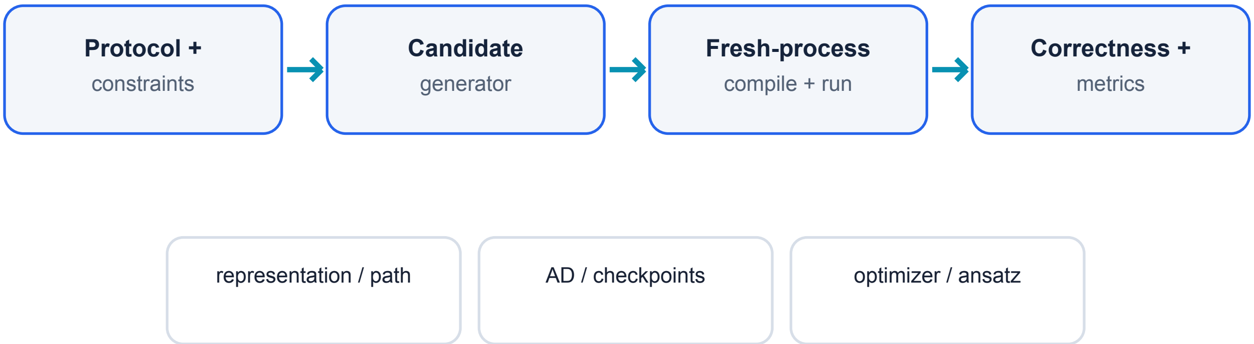
Matched physical protocol

Open-boundary transverse-field Ising model: H = -J sum Z_i Z_(i+1) - g sum X_i, with J=g=1. The matched run uses n=10, depth L=4, the plus initial state, RZZ then RX per layer, seed 33, complex64, and five synchronized warm repetitions on one NVIDIA RTX 3090.

Selection objective

T_objective = T_compile + T_first + 100 x median(T_warm). Compilation includes the measured path-search cost. Correctness requires both energy and the complete gradient to pass 1e-5 tolerances.

Compiler feedback loop



Auto-evaluation feeds measured cost and validity back into selection

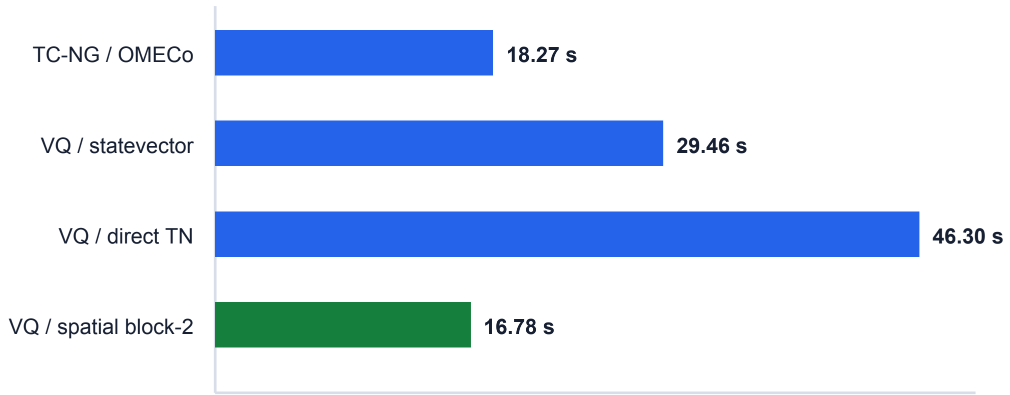
A candidate is eligible only after exact value-gradient validation. Time, host RSS, compiler temporaries, logical residuals, modeled checkpoints, and job-level NVML samples remain separate fields; the report never relabels one as another.

2. Matched performance result

All rows below use the same workload and node. The green row is the VQETape candidate selected for the 100-step objective, not the fastest warm candidate.

Implementation	Compile (s)	First (s)	Warm (ms)	Objective (s)	Host RSS (MiB)	NVML (MiB)
TensorCircuit-NG / OMECo	17.7799	0.2263	2.6577	18.2720	661.3	272
VQETape / statevector	28.7213	0.4025	3.3785	29.4617	462.4	272
VQETape / direct TN	45.6544	0.1376	5.0498	46.2970	575.0	274
VQETape / spatial block-2	15.5816	0.3636	8.3281	16.7781	473.9	274

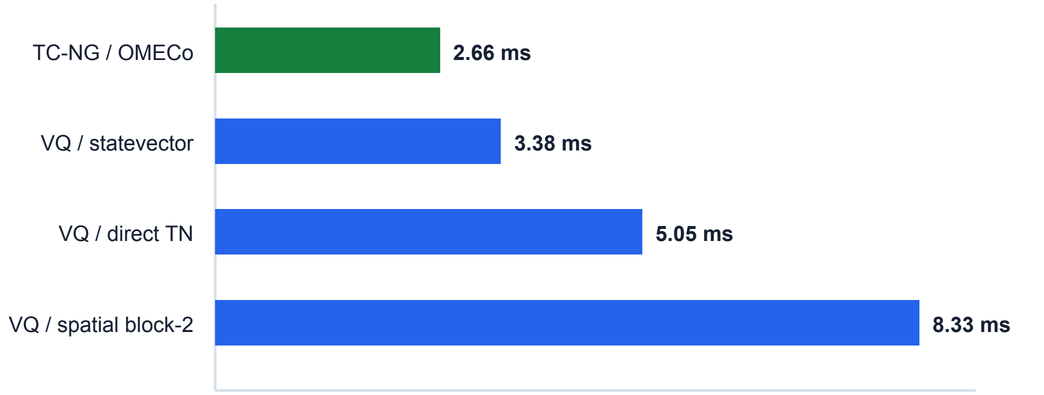
Compile + first + 100 warm (lower is better)



The spatial program crosses the amortized threshold at 16.7781 s versus 18.2720 s. Its advantage comes from compilation: 15.5816 s versus 17.7799 s.

3. Runtime and memory boundary

Warm value-gradient median (lower is better)



Peak host process RSS (lower is better)

Warm runtime is the main unresolved challenge criterion. TensorCircuit-NG records 2.6577 ms; the fastest VQETape warm path is statevector at 3.3785 ms (1.27x slower), while the selected spatial path records 8.3281 ms.

Host RSS is a clear matched win (661.3 to 473.9 MiB). Sampled job-level NVML peaks are 272, 272, 274, and 274 MiB; a 2 MiB spread is treated as tied, not as a device-memory win.

4. Correctness and provenance

The baseline is useful only if it is mathematically and operationally comparable. VQETape therefore records protocol, seed, precision, value, full gradient, timing boundaries, memory semantics, node/job identity, and source checksums.

Gate	Evidence	Result
Matched correctness	Energy + complete gradient; 1e-5 tolerance	PASS
Full regression	395 passed; 6 documented structural skips	PASS
Incremental suite	17 baseline and Fig. 2 tests	PASS
Static artifacts	27 JSON parse; Python compile; diff check	PASS
TC-NG GPU run	Slurm 23020496; cuda:0; SHA256 audit	PASS
Fig. 2 protocol	RTX 3080 N=6,L=3 smoke	PASS / SMALL
Formal Fig. 2	N=32,L=16 on paper-comparable hardware	OPEN

Fig. 2 structural smoke

The separate SU(4) runner preserves 15 x L x (N-1) parameters, TensorNetwork FiniteTFI MPO construction, find/execute separation, slicing controls, and checksum-bound JSON path artifacts. Job 23027373 passed at N=6, L=3 with energy error 2.38e-07 and gradient relative L2 error 3.29e-07.

This proves protocol construction and safe artifact replay. It does not reproduce the H200 N=32,L=16 timing point.

5. Technical contribution

VQETape explores a cross-layer design space instead of treating contraction order as the whole problem.

1 - Exact program representations

Statevector, direct bra-operator-ket tensor network, and spatial transfer with an exact bond-dimension-three TFIM MPO. Gate and Hamiltonian tensors are costed separately.

2 - Differentiated contraction programs

Explicit contraction-tree VJPs, named residuals, checkpoint policies, block widths, scan/unroll choices, and exact Z2 sector compression expose runtime/tape tradeoffs hidden by forward-only FLOP scores.

3 - End-to-end VQE co-design

Optimizer, initialization, recycling, and adaptive ansatz growth include compilation and screening overhead. A commutator-complete YZ/ZY pool reaches 5.05e-11 energy error with 10 parameters; the 14-parameter fixed control stops at 1.70e-07.

4 - Negative results as evidence

Sparse symmetry metadata can cost more than the removed dense carry; exact natural gradient may save iterations but lose wall time; operator-Schmidt gates may reduce logical tape but lose executable runtime; and default GPU matmul precision failed strict correctness until the TensorNetwork backend precision was set explicitly.

The system is an exact 1D research prototype, not a generic Python source-to-source optimizer for arbitrary TensorCircuit-NG scripts.

6. Submission readiness and reproduction

Requirement	Status	Reviewer consequence
Repository and harness	COMPLETE	Installable package, CLI/API, tests, safe artifacts
Controlled TC-NG baseline	COMPLETE	Same node, protocol, seed, precision, correctness
Amortized time	PASS	8.2% faster at matched size
Host memory	PASS	28.3% lower peak RSS
Warm runtime	NOT MET	Best VQETape is 1.27x slower
GPU memory	OPEN	Sampled peaks tied; no superiority claim
Formal Fig. 2 scale	OPEN	Protocol/smoke only

Clean reproduction

```
python3.12 -m venv .venv
.venv/bin/python -m pip install -e '.[test,baseline]'
.venv/bin/python -m pytest -q
python scripts/build_submission_report.py
```

Reviewer entry points

README.md - result and fast-start navigation
submission/vqetape-matched-benchmark.tsv - compact machine-readable rows
submission/submission-status.txt - literal pass/fail boundary
submission/vqetape-technical-report.md - full narrative
submission/report.html - standalone browser report
submission/artifact-manifest.json - SHA256 artifact binding

Final conclusion

The basic engineering and review-delivery requirements are complete, and the project exceeds a minimal baseline wrapper in breadth, correctness discipline, and program co-design. The primary performance challenge remains partially complete until one VQETape path also beats TensorCircuit-NG warm runtime and demonstrates a defensible device-memory advantage.